

# DIT047/DAT357 Requirements Engineering Final Exam

March 18, 2026

## Examiner/Contact Person

Jennifer Horkoff

Jennifer will come to the rooms to check for questions at roughly 14:30, 16:00, and 17:00.

Alternatively Phone: 0733 050517 (call again if no answer, I am in another exam room)

## Authorized Aids

Writing instruments (pens, pencils), English Dictionary. No textbook or notes.

## Grading Scale for Exams

| % Grade | Final Grade |
|---------|-------------|
| 0-49    | Fail (U)    |
| 50-64   | 3           |
| 65-79   | 4           |
| 80-100  | 5           |

PLEASE OBSERVE THE FOLLOWING:

- All answers must be in English. Answers must be legible and readable.
- Sort the questions in order before handing them in.
- Put the number of the question on every page.
- This exam has 12 pages.

## Part 1: Multiple item and Short-Answer Questions

### Question 1: Multiple Choice Questions (15 points)

On your paper, write the name of the question, then the letter(s) of your chosen answer(s), e.g., 1.1: c., 1.3: a, d, e. For each question, there may be more than one correct answer. For full marks, list all the correct answers. Each correct answer listed is +1 point, each incorrect answer is -1 point.

1. Which of these requirements captures design information and is therefore **not** well-formed?  
Note: these requirements focus on a bike rental system. In this question there may be more than one correct answer, list them all.
  - (a) The mobile app shall use a map interface built with Google Maps API to show bike locations.
  - (b) The system shall display bike availability within 5 seconds of a user request.
  - (c) The system shall record the start and end time of each rental.
  - (d) The system shall allow users to find the nearest available bike.
  - (e) The system shall use a three-tier architecture separating UI, business logic, and data storage.
  - (f) The system shall store all rental records in a MySQL relational database

**Answer:** a, e, f

2. Which of the following statements about creative ideas is correct? One or more may be correct.
  - (a) Creative ideas are defined as ideas that are novel and implementable.
  - (b) Creative ideas are defined as ideas that are simple and novel and useful.
  - (c) Creative ideas are defined as ideas that are well-known and useful.
  - (d) All creative ideas become requirements.
  - (e) Creative ideas may be implemented by one or more requirements.
  - (f) Creative ideas are best generated in groups.
  - (g) Creative ideas are best generated alone.

**Answer:** b, e, f

3. When conducting requirements elicitation as part of agile, what challenges commonly hold? One or more may be correct.
  - (a) Stakeholders are too technical.
  - (b) Stakeholders being too busy to participate regularly.
  - (c) Lack of working software.
  - (d) One stakeholder cannot easily represent all stakeholders.
  - (e) Stakeholders feel responsible for requirements.
  - (f) Requirements should not act as legal contracts.

**Answer:** b, d, f

4. Which of the following definitions about safety-related concepts is correct? One or more may be correct.
  - (a) Risk is the probability of occurrence of harm.

- (b) Risk is the combination of the probability of occurrence of harm and the severity of that harm.
- (c) Unreasonable risk is when risk is judged to be unacceptable in a certain context according to valid societal moral concepts.
- (d) Unreasonable risk is the combination of the probability of occurrence of harm and the severity of that harm.
- (e) Safety is the absence of risk.
- (f) Safety is the absence of unreasonable risk.

**Answer:** b, c, f

5. Which statements about security requirements concepts are true? One or more may be correct.
- (a) Security goals state desires.
  - (b) Security requirements state desires.
  - (c) Security goals have a narrow scope.
  - (d) Security requirements have a narrow scope.
  - (e) Security goals are not testable.
  - (f) Security requirements are not testable.

**Answer:** a, d, e

## Question 2: Short-Answer Questions (20 points)

1. List two requirements elicitation techniques. List one advantage and disadvantage of each. (4 point)

**Answer:** See the lecture on elicitation, there are many listed, and each has advantages and disadvantages.

2. These requirements are not verifiable (testable). For each requirements, why are they not verifiable? Rewrite each to be verifiable. (4 points)

1. The system shall provide an easy-to-use interface for renting bikes.
2. The bike rental system shall display bike information.

**Answer:**

1. What does easy to use mean? How easy?

The system interface shall allow all users to complete all tasks within one minute.

The system interface shall be judged as easy to use in a 5-point Likert scale with an average results of 4 or higher for at least 20 users.

The system interface shall be easy to learn, new users shall be able to complete all tasks after 1 hour of training.

2. What information?

The bike rental system shall display bike information including type of bike, weight, and size.

3. In the lectures, we covered several templates and structured natural language approaches for capturing requirements. Name two of them. Name one benefit and one drawback of using a template or structured natural language representation for capturing requirements. (4 points)

**Answer:**

Approaches:

user stories

Ears

Ruppe's template

Planguage

SRS format

Volere

benefits:

Requirements are more uniform

forces people to write complete information

Requirements can be easier to read

Requirements easier to process by tools or AI

drawbacks:

can be harder to write

there is a learning curve

sometimes the templates are not flexible enough, they can not fit what you want to write takes more time

4. Explain, both backwards and forwards requirements traceability. (4 points)

**Answer:**

Backward traceability: Links requirements to their sources which are in e.g. other documents.

Forward: Links requirements to the design and implementation components.

5. In the guest lectures you saw various examples of how different types of AI could be used to improve activities in requirements engineering. Give two examples of how AI can be used to help requirements engineering activities. Be as specific as possible. Describe one drawback of each of these examples. (4 points)

**Answer:** From Farnaz's lecture:

AI can help in requirements elicitation by analyzing stakeholder documentation to extract requirements or helpful information, or analyzing interview or survey text.

AI can help in requirements analysis by detecting ambiguous or inconsistent requirements.

AI can help with requirements prioritization by using data from past projects to estimate priorities of requirements.

AI can help with requirements traceability by automatically finding trace links from or to or between requirements.

AI can help in change management by predicting evolving or changing requirements, or finding what requirements or code or tests are affected by changing requirements.

AI can help in regulatory compliance by seeing which requirements must be added, removed or changed when a regulation changes.

Drawbacks:

AI can hallucinate

Some parts of the output will not be good, will have to be rejected or changed

AI is expensive, has environmental costs

A lot of AI, e.g., chatGPT, breaks privacy by sharing data with model owners, should not be used for confidential data

AI can be hard to set up, depending on the type

Gen AI requires careful prompt design, the output changes a lot depending on the prompts

If we use AI too much, human skill can be lost

## Part 2: Domain Example and Long-Answer Questions

The remaining questions on the exam will relate to a problem in a domain, as described below. The scenario describes the situation today and the expectations for the new system. Focus your analysis and modeling on the envisioned (to be) system, but keep in mind the problems and requests with the as is situation, trying to avoid problems and satisfy user needs.

### Campus Classroom Security

**The Current Situation:** At a university located next to a local high school, classes are often disrupted by high school students who gather outside the classrooms. These students make loud noise in the hallways, bang on the doors and windows, and would sometimes enter classrooms if the doors were not kept locked. This creates an unsafe and distracting environment for university students and faculty, making it difficult to hold lectures and maintain a calm learning atmosphere.

To address this problem, the university currently uses a security guard provided by a private security agency. However, this solution is expensive, and even with a guard present, the high school students still gather nearby and sometimes continue to make noise. When no guard is on site, it can take more than 45 minutes for one to arrive after being called, which leaves classes unprotected during disturbances. Another challenge is that the university has no easy way to keep track of what incidents occur over time. Reports to the university administration, the high school leadership, and the security provider are difficult to prepare because information is scattered, incomplete, or reported via email without a consistent record.

**Vision (System To Be):** The university plans to introduce a new digital system to improve the handling of security incidents and reduce classroom disruptions. If no security guard is present, users should be able to use the system to call one directly. In more serious situations, the system should also offer a fast way to contact the police. Because users may not always know the right response to a disturbance, the system should give clear guidance on what actions to take: when to stay inside the classroom, when to call a guard, and when a situation is serious enough to alert the police.

This system will allow university students, faculty, and security guards to record incidents as they occur, using either a mobile app or a web interface. Video and/or audio can be recorded or uploaded. Related incidents, such as disruptions caused by the same group of students recorded by many users, should be linked together by an administrator so patterns can be understood more clearly.

Over time, the system will store a full history of all incidents and generate statistics that can be used to create summaries and official reports for the university, for the nearby high school, and for the security agency. These reports can help all parties understand the level of disruption, evaluate trends, and decide on long-term actions.

Due to the safety-critical context, the system must be fast and readily available. The system must follow GDPR rules to protect personal data and must meet the Web Content Accessibility Guidelines so it is usable by all members of the university community. The goal is to have the system fully ready and available for use by January 2027.

### Question 3: Context Diagram (10 points)

For the given case, draw a context diagram. Add the system actor, identify the relevant stakeholder/actor entities, and the relationships between the system and these entities, labelled with high-level inputs and outputs. Try to capture four actors (the system actor counts), and appropriate relationships. Supplement the diagram with text to explain any ambiguous or unclear parts of the model.

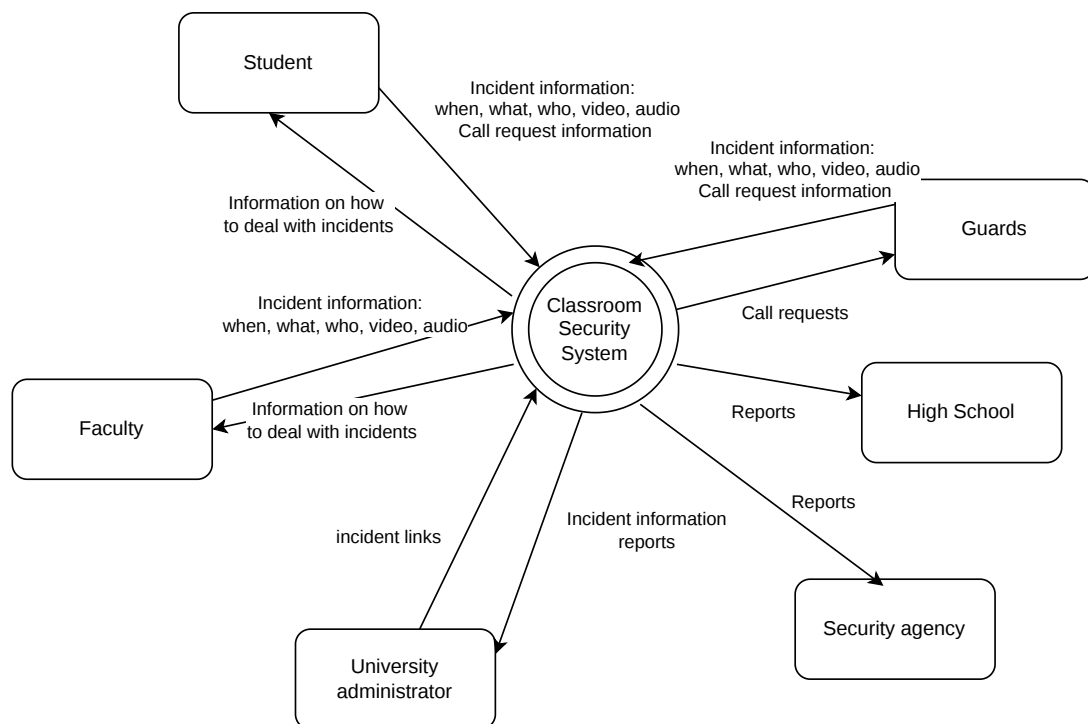
#### Sample Answer:

Grading

4 marks for actors

6 marks for at least 6 correct relationships

A sample answer, many are possible:

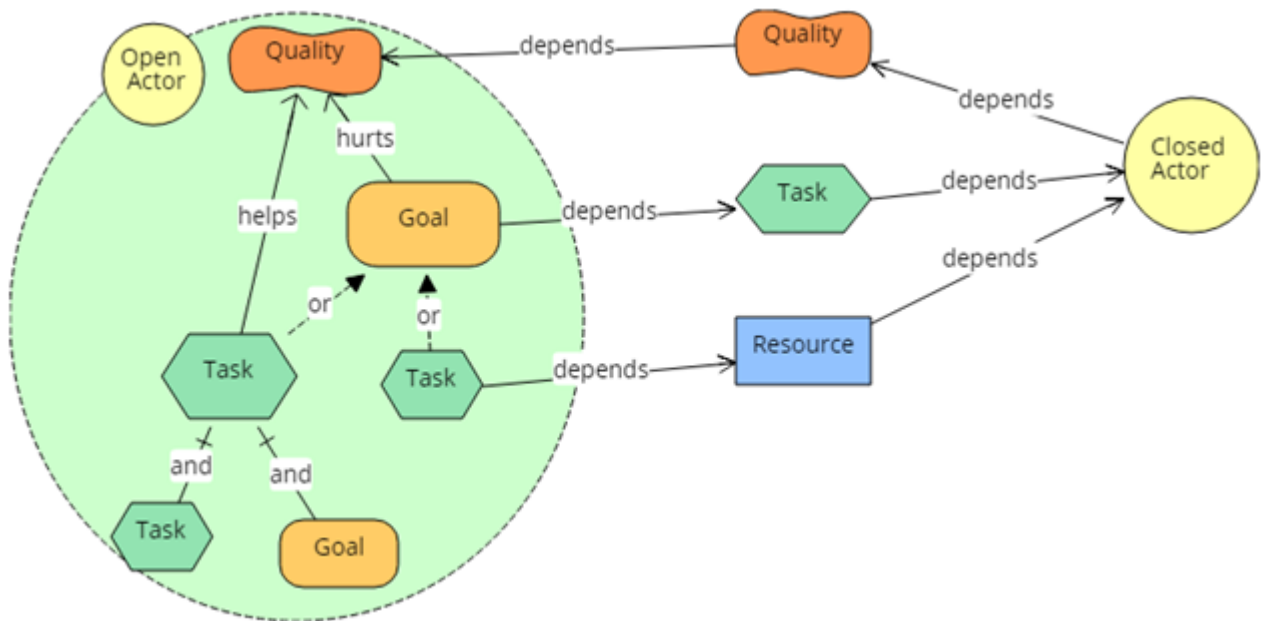


### Question 4: Goal Model (10 points)

For the given case, draw a goal model. Identify the relevant actors, including the system actor. Try to capture at least four actors, and important dependencies between actors. Show the internal goals/tasks/resources/qualities for at least one actor, the others can be "closed" showing only dependencies to and from the actors. See the legend below for open and closed actors. For the open actor, capture the desired goals, tasks, resources, and qualities. Add some internal relationships between these elements.

There are many elements that can be included in the model. Aim for: four actors including one open actor with an actor boundary, at least one goal, quality, resource, task, four dependency links, two contribution links and two AND/OR links. In other words, the model does not have to be complete as per the domain description. It's helpful to use the information already captured in the other diagrams; however, we won't mark you on consistency between diagrams. Supplement the diagram with text to explain any ambiguous or unclear parts of the model.

Goal Model Legend (note, can use alternative shapes and labels as long as the mapping to the original concepts is clear)

**Sample Answer:**

Actors: 2 marks

Goals: 1 mark

Qualities: 1 mark

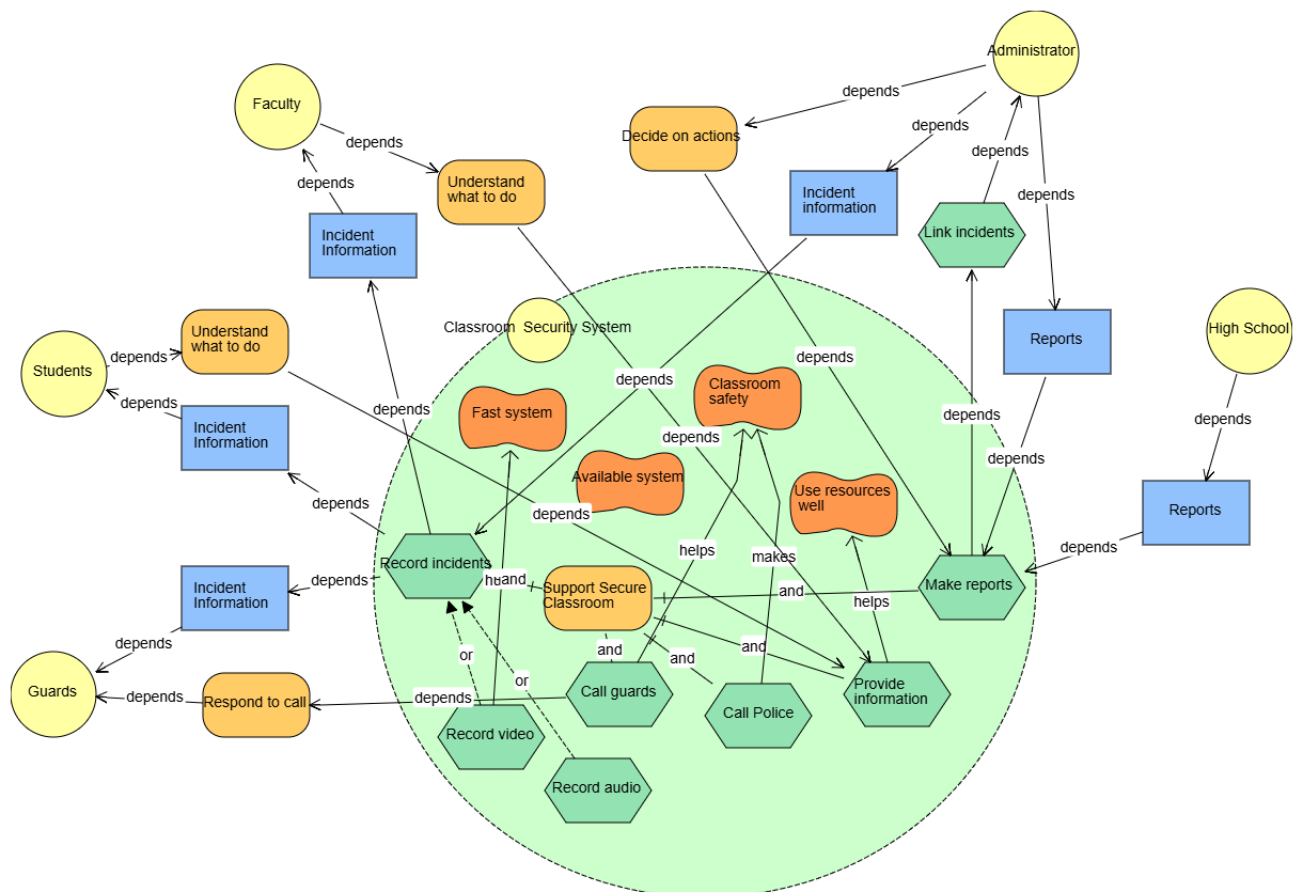
Tasks: 1 mark

Resources: 1 mark

Dependencies: 2 marks

Contributions: 1 mark

AND/OR: 1 mark





**Question 5: Customer Journey Map (10 points)**

Draw a customer journey map for the given case. Indicate the persona that the journey applies to. Think about the channels and touchpoints that the persona would have to go through when using the system. Try to include at least three channels and five touchpoints.

**Sample Answer:**

Persona: 1 mark

Channels: 3 marks

Touchpoints: 5 marks

Timeline: 1 mark

Here are two possible answers

Persona: Jane Smith, student, 23

|             | Go to class | See incident | Record video | Record Incident | Upload video |
|-------------|-------------|--------------|--------------|-----------------|--------------|
| Classroom   | ★           | ★            |              |                 |              |
| App         |             |              |              | ★               | ★            |
| Report      |             |              |              |                 |              |
| Web version |             |              |              |                 |              |
| Phone       |             |              | ★            |                 |              |

Persona: Per Perpersson, faculty member, 50

|             | Give lecture | See incident | Record video | Report Incident | Read report |
|-------------|--------------|--------------|--------------|-----------------|-------------|
| Classroom   | ★            | ★            |              |                 |             |
| App         |              |              | ★            |                 |             |
| Report      |              |              |              |                 | ★           |
| Web version |              |              |              | ★               |             |
| Phone       |              |              |              |                 |             |

**Question 6: Textual Requirements (20 points)**

Write SRS requirements and user stories for the case provided. List four functional requirements in SRS form, four non-functional requirements in SRS form, two domain assumptions, and two constraints. In addition, list four user stories, each with acceptance criteria. The user stories should be unique, i.e. not a repeat of one of the SRS requirements. Remember the desired characteristics of user stories and SRS requirements when writing your requirements.

**Sample Answers:**

Functional requirements

1. The system shall allow university students, faculty, and security guards to record security incidents by entering details such as time, location, description, and people involved.
2. The system shall allow users to call a security guard through the interface when no guard is present.
3. The system shall also provide an option to call the police.
4. The system shall allow the users to upload video or audio files with their incidents in format mp4 or mp3.
5. The system shall allow authorized users (students, faculty, guards, administrators) to view a list of past incidents, filtered by date, location, or incident type.
6. The system shall display guidance to users about what actions to take during different types of disturbances, including when to call a guard and when to contact the police.

NFRs

5. The system shall process and store any incident report within 3 seconds under normal operating conditions.

6. The system shall be available 99% of the time.

7. The system shall ensure that only authorized users (students, faculty, security staff, and administrators) can view or submit incident reports.

8. The system make it easy to call for help, 95% of users in test evaluations should be able to find this function within 5 seconds.

#### Constraints

C1. The system shall comply with the General Data Protection Regulation (GDPR).

C2. The system shall comply with the Web Content Accessibility Guidelines (WCAG).

C3. The system shall be fully operational and available for use by January 2027.

#### Assumptions

We assume all instructors, students, and security guards have phones.

We assume all user phones have data or internet connections.

We assume all students can read and write in English.

We assume all guards and administrators can read and write in English.

We assume someone cares to read the reports and do something about them.

#### User stories

1. As a university student I want to report a disturbance happening outside my classroom so that the university and security staff are aware and can respond quickly.

##### Acceptance Criteria

A user can create an incidence report.

The incident report allows entry of time, location, and a short description.

The incident becomes visible to authorized staff (security and administrators).

The user receives a confirmation message that their report was successfully submitted.

2. As an administrator I want to link multiple incident reports that refer to the same recurring issue so that the university can better track patterns and prepare accurate reports.

##### Acceptance Criteria

The administrator can view all existing incident reports in a list.

The system allows selecting two or more related incident reports.

Linked incidents appear grouped together in the system's interface.

Linked incidents are treated as one group when generating statistics and weekly or monthly reports.

3. As a faculty member I want to quickly call a security guard when a disturbance escalates so that the situation can be handled before it disrupts the class further.

##### Acceptance Criteria

The system displays way to call a guard if no guard is currently present.

The user can to confirm the request before calling.

Once confirmed, the guard receives the user's location and contact information.

The user receives a confirmation that the call was successfully sent.

4. As a faculty member, I want to see reports which cover incidents I have been involved in, so I can understand the bigger picture and see trends.

##### Acceptance Criteria

Faculty members can search for people mentioned in reports to find reports where they are mentioned.

Faculty members can open or download reports from a list

Reports should open or be downloaded within 3 seconds from the request

If no reports are available, a notice should inform the user

If no search results are found for a person, a notice should inform the user

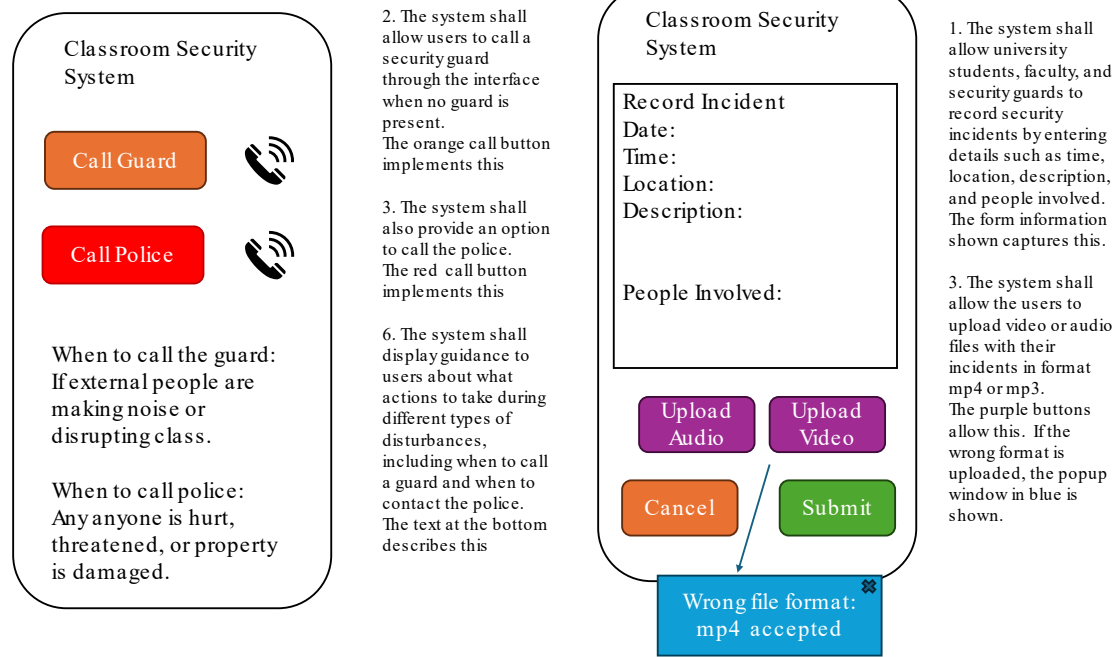
## Question 7: Prototype (10 points)

Draw two prototype screens for the given case implementation. The screens can include pop-up windows, just be clear how and why they appear. The purpose of the screens should be clear. The screen should implement at least two functional requirements (SRS or user stories).

These requirements should be clearly implemented on the screens. Write the full text of the requirements implemented here, so we do not have to flip back to find them. For each screen, write: the requirements implemented, and explain how and exactly where the requirements are implemented.

### Sample Answers:

We give 8 points for the requirements, and 2 points for the screen.



### Implemented Functional Requirements

1. The system shall allow users to request enrollment in available courses.

how and where: The buttons on the far right of the screen associated with each needed course allows teachers to request enrollment for each course.

4. The system shall allow users with default permissions to view the courses that they have completed.

How and where: This is shown in the table on the left under Completed courses after the teacher has logged in.

17. As a teacher, I want to be able to see what courses I need to take, so I can enroll in the courses.

How and where: This is shown by the table of needed courses on the right of the screen.

### Question 8: User Testing Tasks (5 points)

Come up with a list of five test tasks for the users to perform during usability tests for the designed screens. Each screen should be tested by at least two tasks. For these tasks, indicate which of your screens they test. The tasks should cover some of the main use cases or requirements of your system. The role of the user should be clear. Tasks should conform to the desirable qualities of user test tasks as discussed in the lecture.

### Answer:

Sample answers, there are many:

1. (Screen 1) You are student and someone is making noise outside, disrupting the class, what do you do? Can you use the app to help?

2. (Screen 1) You are a faculty member and someone just broke the window of the classroom, what do you do? Can you use the app to help?

3. (Screen 2) You are a faculty member and would like to upload a video of some high school students making noise outside the class. Can you do this in the app?
4. (Screen 2) You are a security guard and have just dealt with an incident involving a class disruption, can you use the app to report the incident?
5. You are an administrator and would like to see all the security incidents in Lindholmen his month, can you get this information from the app?